

Phi-Renormalization for Axisymmetric-with-Swirl Navier–Stokes: A Conditional Reduction of Global Regularity

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Abstract

We study the three-dimensional incompressible Navier–Stokes equations with axisymmetric-with-swirl initial data on $\mathbb{R}^3$, augmented by the linear fractional hyperdissipation $Q_1 = -\varepsilon(-\Delta)^{(\beta+2)/2}$ with $\beta = 0.6 > \frac{1}{2}$. By the Lions threshold [9], the augmented system admits a unique global smooth solution for every fixed $\varepsilon > 0$.

The central technical contribution is the *Phi-renormalization* (Theorem 6.3): the substitution $\Phi = u^\theta/r$, together with $\Gamma = ru^\theta$, yields the identity $r^{-4}\partial_z(\Gamma^2) = \partial_z(\Phi^2)$. This rewrites the swirl-vorticity source term without the explicit $1/r^4$ prefactor, relocating the axis difficulty into the advection coefficient u^r/r ; it does not by itself remove that difficulty.

For the augmented system we establish: (i) uniform-in- ε L^2 energy and Γ -bounds; (ii) a Phi-energy estimate for the renormalized swirl Φ with an explicit Gronwall constant $C(\varepsilon)$ depending on $\|u_\varepsilon^r/r\|_{L^\infty}$; (iii) weak compactness via Aubin–Lions as $\varepsilon \rightarrow 0$; (iv) every subsequential limit is a Leray–Hopf weak solution of the classical system. Whether this weak solution is globally smooth reduces to a single explicitly stated open condition: ε -independence of $C(\varepsilon)$ (Open Problem 10).

What this paper does not prove. Unconditional global regularity of the axisymmetric-with-swirl system: the reduction is conditional on the ε -independent bound of Open Problem 10.

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Contents

1	Introduction	1
2	Notation and Function Spaces	2
3	The Axisymmetric-with-Swirl Equations	3
3.1	Scalar form in cylindrical coordinates	3
3.2	The Γ equation	3
3.3	The vorticity equation for Ω	4
3.4	The equation for Φ	4
3.5	The augmented scalar system	5
4	The Q_1 Augmentation and the Lions Threshold	5
5	Uniform Energy Estimates	5
5.1	L^2 energy inequality	5
5.2	L^∞ bound on Γ	5

6	The Phi-Renormalization	6
6.1	The algebraic cancellation identity	6
6.2	Effect on the vorticity equation	6
7	The Phi-Energy Estimate in Full	7
7.1	Taking the L_r^2 inner product	7
7.2	The advection term T_{adv}	7
7.3	Bounding T_{adv}	7
7.4	The viscous term T_{visc}	7
7.5	Assembling the Phi-energy estimate	8
7.6	Nature of the constant $C(\varepsilon)$	8
8	Global Regularity of the ε-System	8
9	Compactness and the Limit $\varepsilon \rightarrow 0$	9
9.1	Setup	9
9.2	Aubin–Lions compactness	9
9.3	The limit is a Leray–Hopf weak solution	9
10	Open Problems	10
11	Conclusion	11

1 Introduction

The global regularity of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \operatorname{div} u = 0, \quad u(0) = u_0, \quad (1)$$

is one of the seven Clay Millennium Prize Problems [4]. Leray [7] constructed global weak solutions; their smoothness for all time remains unresolved.

In the *axisymmetric-with-swirl* class, the flow has an additional algebraic structure in cylindrical coordinates: the velocity field is independent of the azimuthal angle θ and carries a nonzero swirl component u^θ . The swirl introduces a centrifugal forcing term in the vorticity equation that is potentially singular on the symmetry axis $r = 0$. Controlling this term is the central difficulty of the axisymmetric-with-swirl problem; in the absence of swirl (i.e., $u^\theta \equiv 0$), global regularity is known [6, 13].

We study the Q_1 -augmented system

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u - \varepsilon(-\Delta)^{(\beta+2)/2}u, \quad \operatorname{div} u = 0, \quad u(0) = u_0, \quad (2)$$

with $\varepsilon > 0$ and $\beta = 0.6$. The augmentation is a *linear* fractional hyperdissipation; with $\beta > \frac{1}{2}$, Lions [9] guarantees global smooth solutions for each fixed $\varepsilon > 0$. Our contribution is to analyze the limit $\varepsilon \rightarrow 0$ for axisymmetric-with-swirl data by exploiting the Phi-renormalization.

Organization

Section 2 fixes notation. Section 3 derives the axisymmetric-with-swirl scalar equations. Section 4 records the Lions theorem. Section 5 establishes uniform L^2 and L^∞ bounds. Section 6 contains the Phi-renormalization. Section 7 derives the Phi-energy estimate in full. Section 8 establishes global regularity of the ε -system. Section 9 proves the Aubin–Lions compactness and the limit theorem. Section 10 states the open problems precisely. Section 11 concludes.

2 Notation and Function Spaces

We work on $\mathbb{R}^3$ with cylindrical coordinates (r, θ, z) where $r = \sqrt{x_1^2 + x_2^2} \geq 0$.

Sobolev spaces. For $s \geq 0$, $H^s(\mathbb{R}^3)$ is the standard Sobolev space with

$$\|f\|_{H^s}^2 = \int_{\mathbb{R}^3} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi, \quad \|f\|_{H^s}^2 = \int_{\mathbb{R}^3} |\xi|^{2s} |\hat{f}(\xi)|^2 d\xi.$$

Weighted L^2 spaces. We use the measure $r dr dz$ throughout; explicitly,

$$\|f\|_{L_r^2}^2 = \int_0^\infty \int_{-\infty}^\infty |f(r, z)|^2 r dr dz$$

and H_r^1 is defined with $\|\nabla f\|_{L_r^2}$ as the gradient part of the norm. We also use the measure $dr dz$ (no weight) and write $L^2(dr dz)$ to distinguish.

Axisymmetric-with-swirl fields. A divergence-free velocity field is axisymmetric-with-swirl if

$$u = u^r(r, z, t) \mathbf{e}_r + u^\theta(r, z, t) \mathbf{e}_\theta + u^z(r, z, t) \mathbf{e}_z.$$

Axis regularity conditions: $u^r = u^\theta = \partial_r u^z = 0$ on $r = 0$. These are compatible conditions preserved by the flow.

Key variables.

$\Gamma := r u^\theta$	(angular momentum per unit mass),
$\Phi := u^\theta / r$	(renormalized swirl),
$\omega^\theta := \partial_z u^r - \partial_r u^z$	(azimuthal vorticity),
$\Omega := \omega^\theta / r$	(renormalized azimuthal vorticity).

Note $\Gamma = r^2 \Phi$ (equivalently $u^\theta = r \Phi$), and Φ extends smoothly across $r = 0$ because $u^\theta = O(r)$ at the axis.

Constants. C denotes a positive constant depending only on ν , $\|u_0\|_{L^2}$, and universal Sobolev constants. Dependence on ε is always written $C(\varepsilon)$ and tracked explicitly.

3 The Axisymmetric-with-Swirl Equations

3.1 Scalar form in cylindrical coordinates

For an axisymmetric-with-swirl velocity field the incompressibility condition reduces to

$$\frac{1}{r} \partial_r (r u^r) + \partial_z u^z = 0, \quad \text{equivalently} \quad \partial_r (r u^r) + \partial_z (r u^z) = 0. \quad (3)$$

The θ -component of the Navier–Stokes momentum equation is

$$\partial_t u^\theta + u^r \partial_r u^\theta + u^z \partial_z u^\theta + \frac{u^r u^\theta}{r} = \nu \left(\partial_r^2 u^\theta + \frac{1}{r} \partial_r u^\theta - \frac{u^\theta}{r^2} + \partial_z^2 u^\theta \right). \quad (4)$$

The centripetal term $u^r u^\theta / r$ on the left couples the swirl and radial motion.

3.2 The Γ equation

Lemma 3.1. *The angular momentum $\Gamma = ru^\theta$ satisfies*

$$(\partial_t + u^r \partial_r + u^z \partial_z) \Gamma = \nu \left(\partial_r^2 \Gamma - \frac{1}{r} \partial_r \Gamma + \partial_z^2 \Gamma \right). \quad (5)$$

Proof. Multiply (4) by r . The left side gives

$$r(\partial_t u^\theta + u^r \partial_r u^\theta + u^z \partial_z u^\theta) + u^r u^\theta = (\partial_t + u^r \partial_r + u^z \partial_z)(ru^\theta) = \frac{D\Gamma}{Dt},$$

where $D/Dt = \partial_t + u^r \partial_r + u^z \partial_z$. For the right side, set $\Gamma = ru^\theta$ and compute using $\partial_r \Gamma = u^\theta + r \partial_r u^\theta$, $\partial_{rr} \Gamma = 2\partial_r u^\theta + r \partial_{rr} u^\theta$: The θ -component of the *vector* Laplacian is $\Delta_{\text{cyl}} u^\theta - u^\theta/r^2$, where $\Delta_{\text{cyl}} = \partial_{rr} + \frac{1}{r} \partial_r + \partial_{zz}$ is the scalar Laplacian acting on the θ -independent profile u^θ and $-u^\theta/r^2$ is the single curvature term. Thus

$$\begin{aligned} \nu r \left(\Delta_{\text{cyl}} u^\theta - \frac{u^\theta}{r^2} \right) &= \nu r \left(\partial_{rr} u^\theta + \frac{1}{r} \partial_r u^\theta + \partial_{zz} u^\theta - \frac{u^\theta}{r^2} \right) \\ &= \nu \left(r \partial_{rr} u^\theta + \partial_r u^\theta + r \partial_{zz} u^\theta - \frac{u^\theta}{r} \right). \end{aligned}$$

Substituting $r \partial_{rr} u^\theta = \partial_{rr} \Gamma - 2\partial_r u^\theta$, $\partial_r u^\theta = \partial_r \Gamma / r - \Gamma / r^2$, and $u^\theta = \Gamma / r$:

$$\begin{aligned} &= \nu \left(\partial_{rr} \Gamma - 2\partial_r u^\theta + \partial_r u^\theta - \frac{u^\theta}{r} + \partial_{zz} \Gamma \right) = \nu \left(\partial_{rr} \Gamma - \partial_r u^\theta - \frac{u^\theta}{r} + \partial_{zz} \Gamma \right) \\ &= \nu \left(\partial_{rr} \Gamma - \left(\frac{\partial_r \Gamma}{r} - \frac{\Gamma}{r^2} \right) - \frac{\Gamma}{r^2} + \partial_{zz} \Gamma \right) = \nu \left(\partial_{rr} \Gamma - \frac{1}{r} \partial_r \Gamma + \partial_{zz} \Gamma \right). \end{aligned}$$

The two curvature contributions cancel exactly, $+\Gamma/r^2 - \Gamma/r^2 = 0$, leaving no Γ/r^2 term and yielding (5). (See also [10], Ch. 2.) $\square$

Remark 3.2. The operator on the right of (5) is $\mathcal{L}_0 = \partial_{rr} - (1/r)\partial_r + \partial_{zz}$. In the weighted space L_r^2 , integration by parts gives

$$\int \mathcal{L}_0 \Gamma \cdot \Gamma \cdot r \, dr \, dz = -\|\nabla \Gamma\|_{L_r^2}^2,$$

so $\mathcal{L}_0$ is dissipative in L_r^2 . The $-(1/r)\partial_r$ term is a drift pointing away from the axis, which helps maintain coercivity.

3.3 The vorticity equation for Ω

Lemma 3.3. *The renormalized azimuthal vorticity $\Omega = \omega^\theta / r$ satisfies*

$$(\partial_t + u^r \partial_r + u^z \partial_z) \Omega = \frac{1}{r^4} \partial_z (\Gamma^2) + \nu \left(\partial_r^2 \Omega + \frac{3}{r} \partial_r \Omega + \partial_z^2 \Omega \right). \quad (6)$$

Proof. The azimuthal component of the NS vorticity equation $\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta_{\text{vec}} \omega$ reduces for axisymmetric fields to

$$(\partial_t + u^r \partial_r + u^z \partial_z) \omega^\theta - \frac{u^r \omega^\theta}{r} = \frac{\partial_z (u^\theta)^2}{r} + \nu \left(\Delta \omega^\theta - \frac{\omega^\theta}{r^2} \right).$$

Divide by r and write $\omega^\theta = r\Omega$. The left side becomes $(\partial_t + u^r \partial_r + u^z \partial_z) \Omega$ (the u^r/r terms cancel). The source term: $(u^\theta)^2 = \Gamma^2/r^2$, so $\partial_z (u^\theta)^2/r^2 = \partial_z (\Gamma^2/r^2)/r^2 = (1/r^4) \partial_z (\Gamma^2)$ (since ∂_z holds r fixed).

For the diffusion term, compute $\nu r^{-1}(\Delta(r\Omega) - \Omega)$:

$$\begin{aligned}\Delta(r\Omega) &= \partial_{rr}(r\Omega) + \frac{1}{r}\partial_r(r\Omega) + \partial_{zz}(r\Omega) \\ &= (2\partial_r\Omega + r\partial_{rr}\Omega) + \left(\frac{\Omega}{r} + \partial_r\Omega\right) + r\partial_{zz}\Omega.\end{aligned}$$

Thus $\Delta(r\Omega) - \Omega = r\partial_{rr}\Omega + 3\partial_r\Omega + \Omega/r + r\partial_{zz}\Omega - \Omega$. Since $\Omega = O(r^0)$ at the axis, the Ω/r and Ω terms combine as $(1/r - 1)\Omega \rightarrow 0$; more precisely, dividing by r :

$$\frac{1}{r}(\Delta(r\Omega) - \Omega) = \partial_{rr}\Omega + \frac{3}{r}\partial_r\Omega + \partial_{zz}\Omega,$$

yielding (6). □

3.4 The equation for Φ

Lemma 3.4. *The renormalized swirl $\Phi = u^\theta/r$ satisfies*

$$\frac{D\Phi}{Dt} + \frac{2u^r\Phi}{r} = \nu \left(\partial_{rr}\Phi + \frac{3}{r}\partial_r\Phi + \partial_{zz}\Phi \right), \quad (7)$$

where $D/Dt = \partial_t + u^r\partial_r + u^z\partial_z$.

Proof. Write $\Gamma = r^2\Phi$. Then

$$\frac{D\Gamma}{Dt} = r^2 \frac{D\Phi}{Dt} + u^r\partial_r(r^2)\Phi = r^2 \frac{D\Phi}{Dt} + 2ru^r\Phi.$$

Substituting $\Gamma = r^2\Phi$ into (5):

$$\begin{aligned}\partial_{rr}(r^2\Phi) &= 2\Phi + 4r\partial_r\Phi + r^2\partial_{rr}\Phi, \\ \frac{1}{r}\partial_r(r^2\Phi) &= 2\Phi + r\partial_r\Phi, \\ \partial_{zz}(r^2\Phi) &= r^2\partial_{zz}\Phi.\end{aligned}$$

So $\nu\mathcal{L}_0\Gamma = \nu(2\Phi + 4r\partial_r\Phi + r^2\partial_{rr}\Phi - 2\Phi - r\partial_r\Phi + r^2\partial_{zz}\Phi) = \nu r^2(\partial_{rr}\Phi + \frac{3}{r}\partial_r\Phi + \partial_{zz}\Phi)$, using that (5) carries no Γ/r^2 term. Setting $r^2 \frac{D\Phi}{Dt} + 2ru^r\Phi = \nu r^2(\partial_{rr}\Phi + (3/r)\partial_r\Phi + \partial_{zz}\Phi)$ and dividing by r^2 gives (7). □

3.5 The augmented scalar system

Including the Q_1 fractional dissipation, the augmented equations for $(\Omega_\varepsilon, \Gamma_\varepsilon, \Phi_\varepsilon)$ are:

$$(\partial_t + u_\varepsilon \cdot \nabla)\Omega_\varepsilon = \frac{1}{r^4}\partial_z(\Gamma_\varepsilon^2) + \nu\mathcal{L}_4\Omega_\varepsilon - \varepsilon(-\Delta)^{1.3}\Omega_\varepsilon, \quad (8)$$

$$(\partial_t + u_\varepsilon \cdot \nabla)\Gamma_\varepsilon = \nu\mathcal{L}_0\Gamma_\varepsilon - \varepsilon(-\Delta)^{1.3}\Gamma_\varepsilon, \quad (9)$$

$$\frac{D\Phi_\varepsilon}{Dt} + \frac{2u_\varepsilon^r\Phi_\varepsilon}{r} = \nu\mathcal{L}_4\Phi_\varepsilon - \varepsilon(-\Delta)^{1.3}\Phi_\varepsilon, \quad (10)$$

where $\mathcal{L}_4 = \partial_{rr} + (3/r)\partial_r + \partial_{zz}$ is the same operator that governs Ω in (6).

4 The Q_1 Augmentation and the Lions Threshold

Theorem 4.1 (Lions 1969, [9]). *Let $\beta \geq \frac{1}{2}$. For each $\varepsilon > 0$ and axisymmetric-with-swirl $u_0 \in H_{\text{axi}}^s(\mathbb{R}^3)$ with s sufficiently large, the Q_1 -augmented system (2) admits a unique global smooth solution $u_\varepsilon \in C^\infty(\mathbb{R}^3 \times [0, \infty))$.*

Our value $\beta = 0.6$ satisfies $\beta > \frac{1}{2}$ strictly. Theorem 4.1 is applied as a black box; the contribution of this paper is the analysis of the limit $\varepsilon \rightarrow 0$ for axisymmetric-with-swirl data via the Phi-renormalization.

5 Uniform Energy Estimates

5.1 L^2 energy inequality

Proposition 5.1. *The Q_1 -augmented solutions satisfy*

$$\frac{1}{2} \frac{d}{dt} \|u_\varepsilon\|_{L^2}^2 + \nu \|\nabla u_\varepsilon\|_{L^2}^2 + \varepsilon \|u_\varepsilon\|_{\dot{H}^{1.3}}^2 = 0. \quad (11)$$

Consequently, $\|u_\varepsilon(t)\|_{L^2} \leq \|u_0\|_{L^2} =: M$ for all $t \geq 0$ and all $\varepsilon > 0$, and

$$\int_0^T \|\nabla u_\varepsilon\|_{L^2}^2 dt \leq \frac{M^2}{2\nu}, \quad \varepsilon \int_0^T \|u_\varepsilon\|_{\dot{H}^{1.3}}^2 dt \leq \frac{M^2}{2}, \quad (12)$$

both uniform in $\varepsilon > 0$.

Proof. Take the $L^2(\mathbb{R}^3)$ inner product of (2) with u_ε . The pressure term vanishes by $\operatorname{div} u_\varepsilon = 0$. The nonlinear term vanishes: $\langle (u_\varepsilon \cdot \nabla) u_\varepsilon, u_\varepsilon \rangle = \frac{1}{2} \int u_\varepsilon \cdot \nabla |u_\varepsilon|^2 = -\frac{1}{2} \int (\operatorname{div} u_\varepsilon) |u_\varepsilon|^2 = 0$. The viscous term gives $-\nu \|\nabla u_\varepsilon\|_{L^2}^2$, and the Q_1 term gives $-\varepsilon \|u_\varepsilon\|_{\dot{H}^{1.3}}^2 \leq 0$. This yields (11). Integrating in t and using non-negativity of all terms gives (12). $\square$

5.2 L^∞ bound on Γ

Lemma 5.2. *If $\Gamma_0 = r u_0^\theta \in L^\infty \cap L^2(\mathbb{R}^3)$, then*

$$\|\Gamma_\varepsilon(t)\|_{L^\infty} \leq \|\Gamma_0\|_{L^\infty} =: G_0 \quad \text{for all } t \geq 0, \varepsilon > 0. \quad (13)$$

Proof. Equation (9) is $D\Gamma_\varepsilon/Dt = \nu \mathcal{L}_0 \Gamma_\varepsilon - \varepsilon (-\Delta)^{1.3} \Gamma_\varepsilon$. Both $\nu \mathcal{L}_0$ and $-\varepsilon (-\Delta)^{1.3}$ are dissipative operators. Let Γ_ε achieve its spatial maximum at (r_0, z_0, t_0) with $r_0 > 0$. At that point $\nabla \Gamma_\varepsilon = 0$ and $\partial_{rr} \Gamma_\varepsilon \leq 0$, $\partial_{zz} \Gamma_\varepsilon \leq 0$, giving $\mathcal{L}_0 \Gamma_\varepsilon \leq 0$. The nonlocal term $-\varepsilon (-\Delta)^{1.3} \Gamma_\varepsilon \leq 0$ at a global maximum by the nonlocal maximum principle [12]. Hence $D\Gamma_\varepsilon/Dt \leq 0$ at the maximum, so the maximum is non-increasing. On the axis $r = 0$, $\Gamma_\varepsilon = r u_\varepsilon^\theta = 0$ by the axis regularity condition. Therefore $\|\Gamma_\varepsilon(t)\|_{L^\infty}$ is non-increasing in t , giving (13). $\square$

6 The Phi-Renormalization

6.1 The algebraic cancellation identity

Lemma 6.1 (Exact algebraic cancellation). *With $\Gamma = r u^\theta$ and $\Phi = u^\theta/r$, so that $\Gamma = r^2 \Phi$, the following identity holds for all (r, z, t) :*

$$\frac{1}{r^4} \partial_z (\Gamma^2) = \partial_z (\Phi^2). \quad (14)$$

In particular, the right-hand side is smooth on the axis $r = 0$, since $\Phi = u^\theta/r$ extends smoothly there (the axis condition $u^\theta = O(r)$ is preserved by the flow). No Hardy inequality is required or used.

Proof. Direct computation: $\Gamma^2 = (r^2 \Phi)^2 = r^4 \Phi^2$. Differentiating in z with r held fixed: $\partial_z (\Gamma^2) = r^4 \partial_z (\Phi^2)$. Dividing both sides by r^4 gives (14). $\square$

Remark 6.2. Identity (14) is purely algebraic and holds independently of whether the flow is viscous, inviscid, or augmented. It is the sole mechanism behind the cancellation; no functional inequality is invoked.

6.2 Effect on the vorticity equation

Substituting (14) into (8), the augmented Ω -equation becomes

$$(\partial_t + u_\varepsilon \cdot \nabla) \Omega_\varepsilon = \partial_z(\Phi_\varepsilon^2) + \nu \mathcal{L}_4 \Omega_\varepsilon - \varepsilon(-\Delta)^{1.3} \Omega_\varepsilon. \quad (15)$$

The source term is now $\partial_z(\Phi_\varepsilon^2) = 2\Phi_\varepsilon \partial_z \Phi_\varepsilon$, which is smooth across $r = 0$.

Theorem 6.3 (Phi-renormalization). *For axisymmetric-with-swirl Q_1 -augmented solutions with $\Gamma_0 \in L^\infty \cap L^2$ and $\Phi_0 = u_0^\theta/r \in L_r^2$:*

- (i) (No axis singularity.) *The singular term $r^{-4} \partial_z(\Gamma_\varepsilon^2)$ in the Ω -equation is identically equal to $\partial_z(\Phi_\varepsilon^2)$ by Lemma 6.1. No residual singularity at $r = 0$ remains.*
- (ii) (Phi-energy estimate.) *There exists a constant $C(\varepsilon)$, bounded for each fixed $\varepsilon > 0$ and depending on $\|u_\varepsilon^r/r\|_{L_{t,x}^\infty}$, G_0 , and M , such that*

$$\frac{1}{2} \frac{d}{dt} \|\Phi_\varepsilon\|_{L_r^2}^2 + \nu \|\nabla \Phi_\varepsilon\|_{L_r^2}^2 + \nu \int_{\mathbb{R}} \Phi_\varepsilon(0, z)^2 dz + \varepsilon \|\Phi_\varepsilon\|_{\dot{H}_r^{1.3}}^2 \leq C(\varepsilon) \|\Phi_\varepsilon\|_{L_r^2}^2. \quad (16)$$

- (iii) (Gronwall bound.) *By Gronwall's inequality,*

$$\|\Phi_\varepsilon(t)\|_{L_r^2} \leq \|\Phi_0\|_{L_r^2} e^{C(\varepsilon)t} \quad \text{for all } t \geq 0. \quad (17)$$

The open condition. *If $C(\varepsilon)$ admits an ε -independent upper bound $C_0 = C_0(M, G_0)$, independent of $\varepsilon > 0$, then the Leray–Hopf weak solution of Theorem 9.2 is globally smooth. This is Open Problem 10.*

The proof of Theorem 6.3 is given in full in the next section.

7 The Phi-Energy Estimate in Full

7.1 Taking the L_r^2 inner product

Take the L_r^2 inner product of (10) with Φ_ε , i.e., multiply by Φ_ε and integrate against $r dr dz$:

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \|\Phi_\varepsilon\|_{L_r^2}^2 + \underbrace{\int \left(u_\varepsilon^r \partial_r \Phi_\varepsilon + u_\varepsilon^z \partial_z \Phi_\varepsilon + \frac{2u_\varepsilon^r \Phi_\varepsilon}{r} \right) \Phi_\varepsilon r dr dz}_{=: T_{\text{adv}}} \\ = \nu \underbrace{\int \mathcal{L}_4 \Phi_\varepsilon \cdot \Phi_\varepsilon r dr dz}_{=: T_{\text{visc}}} - \varepsilon \|\Phi_\varepsilon\|_{\dot{H}_r^{1.3}}^2. \end{aligned} \quad (18)$$

7.2 The advection term T_{adv}

Lemma 7.1. $T_{\text{adv}} = 2 \int u_\varepsilon^r \Phi_\varepsilon^2 dr dz$.

Proof. Write $T_{\text{adv}} = \frac{1}{2} \int u_\varepsilon \cdot \nabla(\Phi_\varepsilon^2) \cdot r dr dz + 2 \int u_\varepsilon^r \Phi_\varepsilon^2 dr dz$. Integrate by parts:

$$\begin{aligned} \frac{1}{2} \int u_\varepsilon^r \partial_r(\Phi_\varepsilon^2) \cdot r dr dz &= -\frac{1}{2} \int \Phi_\varepsilon^2 \partial_r(r u_\varepsilon^r) dr dz, \\ \frac{1}{2} \int u_\varepsilon^z \partial_z(\Phi_\varepsilon^2) \cdot r dr dz &= -\frac{1}{2} \int \Phi_\varepsilon^2 \partial_z(r u_\varepsilon^z) dr dz. \end{aligned}$$

By axisymmetric incompressibility (3), $\partial_r(r u_\varepsilon^r) + \partial_z(r u_\varepsilon^z) = 0$. Hence the two integration-by-parts terms sum to zero, leaving $T_{\text{adv}} = 2 \int u_\varepsilon^r \Phi_\varepsilon^2 dr dz$. $\square$

7.3 Bounding T_{adv}

Factor out u_ε^r/r :

$$|T_{\text{adv}}| = 2 \left| \int \frac{u_\varepsilon^r}{r} \Phi_\varepsilon^2 r \, dr \, dz \right| \leq 2 \left\| \frac{u_\varepsilon^r}{r} \right\|_{L^\infty} \|\Phi_\varepsilon\|_{L_r^2}^2. \quad (19)$$

For smooth axisymmetric fields, the axis regularity condition $u^r|_{r=0} = 0$ together with $u^r = O(r)$ implies u^r/r is bounded and smooth. For the Q_1 -augmented system, Theorem 4.1 ensures $u_\varepsilon \in C^\infty$ for each $\varepsilon > 0$, so

$$C(\varepsilon) := 2 \sup_{t \in [0, T]} \left\| \frac{u_\varepsilon^r(t)}{r} \right\|_{L^\infty} < \infty \quad \text{for each fixed } \varepsilon > 0. \quad (20)$$

Equation (19) becomes $|T_{\text{adv}}| \leq C(\varepsilon) \|\Phi_\varepsilon\|_{L_r^2}^2$.

7.4 The viscous term T_{visc}

Lemma 7.2. $T_{\text{visc}} = -\|\nabla \Phi_\varepsilon\|_{L_r^2}^2 - \int_{\mathbb{R}} \Phi_\varepsilon(0, z)^2 \, dz \leq 0$.

Proof. Recall $\mathcal{L}_4 = \partial_{rr} + (3/r)\partial_r + \partial_{zz}$. We compute each piece against the weight $r \, dr \, dz$.

The ∂_{rr} piece. Integrating by parts in r ,

$$\int \partial_{rr} \Phi_\varepsilon \cdot \Phi_\varepsilon \cdot r \, dr \, dz = - \int (\partial_r \Phi_\varepsilon)^2 r \, dr \, dz - \int \partial_r \Phi_\varepsilon \cdot \Phi_\varepsilon \, dr \, dz,$$

the boundary contribution $[r \partial_r \Phi_\varepsilon \cdot \Phi_\varepsilon]_{r=0}^{r=\infty}$ vanishing (the weight r kills the axis endpoint, decay kills $r = \infty$).

The $(3/r)\partial_r$ piece.

$$3 \int \frac{1}{r} \partial_r \Phi_\varepsilon \cdot \Phi_\varepsilon \cdot r \, dr \, dz = 3 \int \partial_r \Phi_\varepsilon \cdot \Phi_\varepsilon \, dr \, dz.$$

The ∂_{zz} piece.

$$\int \partial_{zz} \Phi_\varepsilon \cdot \Phi_\varepsilon \cdot r \, dr \, dz = - \int (\partial_z \Phi_\varepsilon)^2 r \, dr \, dz.$$

Summing the three pieces,

$$T_{\text{visc}} = -\|\nabla \Phi_\varepsilon\|_{L_r^2}^2 + 2 \int \partial_r \Phi_\varepsilon \cdot \Phi_\varepsilon \, dr \, dz.$$

The remaining term is a genuine axis boundary term and does *not* vanish in general:

$$2 \int \partial_r \Phi_\varepsilon \cdot \Phi_\varepsilon \, dr \, dz = \int_{\mathbb{R}} [\Phi_\varepsilon^2]_{r=0}^{r=\infty} \, dz = - \int_{\mathbb{R}} \Phi_\varepsilon(0, z)^2 \, dz,$$

since $\Phi_\varepsilon \rightarrow 0$ as $r \rightarrow \infty$ while $\Phi_\varepsilon(0, z) = \partial_r u_\varepsilon^\theta|_{r=0}$ need not vanish. Hence

$$T_{\text{visc}} = -\|\nabla \Phi_\varepsilon\|_{L_r^2}^2 - \int_{\mathbb{R}} \Phi_\varepsilon(0, z)^2 \, dz \leq 0,$$

both terms being non-positive. The axis term is *favorable*—it supplies additional dissipation—so the estimate only improves. $\square$

7.5 Assembling the Phi-energy estimate

Substituting Lemmas 7.1 and 7.2 into (18):

$$\frac{1}{2} \frac{d}{dt} \|\Phi_\varepsilon\|_{L_r^2}^2 + \nu \|\nabla \Phi_\varepsilon\|_{L_r^2}^2 + \nu \int_{\mathbb{R}} \Phi_\varepsilon(0, z)^2 dz + \varepsilon \|\Phi_\varepsilon\|_{\dot{H}_r^{1.3}}^2 \leq C(\varepsilon) \|\Phi_\varepsilon\|_{L_r^2}^2. \quad (21)$$

Gronwall's lemma applied to $\frac{d}{dt} \|\Phi_\varepsilon\|_{L_r^2}^2 \leq 2C(\varepsilon) \|\Phi_\varepsilon\|_{L_r^2}^2$ gives

$$\|\Phi_\varepsilon(t)\|_{L_r^2}^2 \leq \|\Phi_0\|_{L_r^2}^2 e^{2C(\varepsilon)t},$$

completing the proof of Theorem 6.3.

7.6 Nature of the constant $C(\varepsilon)$

The constant $C(\varepsilon) = 2 \sup_t \|u_\varepsilon^r/r\|_{L^\infty}$ depends on the H^s regularity of u_ε for $s \geq 2$, which is controlled by the Lions theorem for each fixed $\varepsilon > 0$ but may grow as $\varepsilon \rightarrow 0$. Controlling $\|u^r/r\|_{L^\infty}$ uniformly in ε is precisely the open condition: it is related to Prodi–Serrin type criteria for Navier–Stokes regularity [11]. The connection is made explicit in Open Problem 10.

8 Global Regularity of the ε -System

Theorem 8.1 (Global smooth solutions for NS_ε). *Let $\beta = 0.6$ and $\varepsilon > 0$. For axisymmetric-with-swirl initial data $u_0 \in H_{\text{axi}}^\infty(\mathbb{R}^3)$ with $\Gamma_0 \in L^\infty$ and $\Phi_0 \in L_r^2$, the Q_1 -augmented system (2) admits a unique global smooth solution $u_\varepsilon \in C^\infty(\mathbb{R}^3 \times [0, \infty))$ satisfying:*

- $\|u_\varepsilon(t)\|_{L^2} \leq M$ (uniform in ε and t),
- $\int_0^T \|\nabla u_\varepsilon\|_{L^2}^2 dt \leq M^2/(2\nu)$ (uniform in ε),
- $\|\Gamma_\varepsilon(t)\|_{L^\infty} \leq G_0$ (uniform in ε and t),
- $\|\Phi_\varepsilon(t)\|_{L_r^2} \leq \|\Phi_0\|_{L_r^2} e^{C(\varepsilon)t}$ (for each fixed $\varepsilon > 0$).

Proof. Global existence and smoothness follow from Theorem 4.1 ($\beta > \frac{1}{2}$). The uniform L^2 bound is Proposition 5.1. The Γ -bound is Lemma 5.2. The Φ -bound is Theorem 6.3(iii).

Higher Sobolev bounds follow by a standard elliptic bootstrap: given H^k control on $(\Omega_\varepsilon, \Gamma_\varepsilon, \Phi_\varepsilon)$, the equations (8)–(10) provide H^{k+1} bounds by differentiating the equations and applying the same energy method, with constants depending on ε through $\|u_\varepsilon\|_{H^k}$. The axisymmetric-with-swirl structure is preserved: the axis conditions $u^r = u^\theta = 0$ at $r = 0$ are consequences of the symmetry class and the smoothness of u_ε , and propagate to all times. $\square$

9 Compactness and the Limit $\varepsilon \rightarrow 0$

9.1 Setup

Fix $T > 0$. The family $\{u_\varepsilon\}_{0 < \varepsilon \leq 1}$ satisfies:

$$(U1) \quad \sup_{t \in [0, T]} \|u_\varepsilon(t)\|_{L^2} \leq M, \quad (22)$$

$$(U2) \quad \int_0^T \|\nabla u_\varepsilon\|_{L^2}^2 dt \leq M^2/(2\nu), \quad (23)$$

both uniform in ε .

9.2 Aubin–Lions compactness

Lemma 9.1 (Aubin–Lions). *The family $\{u_\varepsilon\}$ is precompact in $L^2([0, T]; L^2_{\text{loc}}(\mathbb{R}^3))$.*

Proof. From (U1)–(U2), $\{u_\varepsilon\}$ is bounded in $L^2([0, T]; H^1_{\text{axi}}(\mathbb{R}^3))$.

For the time derivative: testing (2) against $\phi \in H^1_{\text{axi}}$ with $\|\phi\|_{H^1} \leq 1$,

$$\begin{aligned} |\langle \partial_t u_\varepsilon, \phi \rangle| &\leq |\langle (u_\varepsilon \cdot \nabla) u_\varepsilon, \phi \rangle| + \nu |\langle \nabla u_\varepsilon, \nabla \phi \rangle| + \varepsilon |\langle (-\Delta)^{1.3} u_\varepsilon, \phi \rangle| \\ &\leq \|u_\varepsilon\|_{L^4}^2 \|\phi\|_{H^1} + \nu \|\nabla u_\varepsilon\|_{L^2} \|\phi\|_{H^1} + \varepsilon \|u_\varepsilon\|_{\dot{H}^{1.3}} \|\phi\|_{\dot{H}^{1.3}}. \end{aligned}$$

Integrating in t and using (U1)–(U2) and Sobolev embedding $H^1 \hookrightarrow L^4$ in $\mathbb{R}^3$: $\{\partial_t u_\varepsilon\}$ is bounded in $L^{4/3}([0, T]; H^{-1}_{\text{axi}})$.

By the Aubin–Lions lemma [1, 9], a bounded set in $L^2([0, T]; H^1) \cap W^{1,4/3}([0, T]; H^{-1})$ is precompact in $L^2([0, T]; L^2_{\text{loc}})$. $\square$

9.3 The limit is a Leray–Hopf weak solution

Theorem 9.2 (Weak limit). *There exists a subsequence $\varepsilon_k \rightarrow 0$ and a divergence-free axisymmetric-with-swirl vector field $u \in L^\infty([0, T]; L^2) \cap L^2([0, T]; H^1)$ such that:*

- (i) $u_{\varepsilon_k} \rightarrow u$ strongly in $L^2([0, T]; L^2_{\text{loc}}(\mathbb{R}^3))$,
- (ii) u is a Leray–Hopf weak solution of the classical system (1) with initial data u_0 .

Proof. Step 1: Extraction. By Lemma 9.1, extract a subsequence $u_{\varepsilon_k} \rightarrow u$ strongly in $L^2([0, T]; L^2_{\text{loc}})$ and weakly in $L^2([0, T]; H^1)$.

Step 2: Limit of the nonlinear term. For $\phi \in C_0^\infty(\mathbb{R}^3 \times [0, T])$:

$$\begin{aligned} \int_0^T \langle (u_{\varepsilon_k} \cdot \nabla) u_{\varepsilon_k}, \phi \rangle dt - \int_0^T \langle (u \cdot \nabla) u, \phi \rangle dt &= \int_0^T \langle (u_{\varepsilon_k} - u) \cdot \nabla u_{\varepsilon_k}, \phi \rangle dt \\ &\quad + \int_0^T \langle u \cdot \nabla (u_{\varepsilon_k} - u), \phi \rangle dt. \end{aligned}$$

The first term: $\|(u_{\varepsilon_k} - u) \cdot \nabla u_{\varepsilon_k}\|_{L^1_{t,x}} \leq \|u_{\varepsilon_k} - u\|_{L^2_t L^4_x} \|\nabla u_{\varepsilon_k}\|_{L^2_{t,x}} \rightarrow 0$ by strong convergence in $L^2_t L^4_x$ (from strong L^2_{loc} and the Sobolev embedding) and (U2). The second term goes to zero by weak convergence of u_{ε_k} in $L^2_t H^1$.

Step 3: Limit of the Q_1 term. For test function $\phi \in C_0^\infty$, the Cauchy–Schwarz inequality and (12) give:

$$\begin{aligned} \left| \int_0^T \varepsilon_k \langle (-\Delta)^{1.3} u_{\varepsilon_k}, \phi \rangle dt \right| &\leq \varepsilon_k^{1/2} \left(\varepsilon_k \int_0^T \|u_{\varepsilon_k}\|_{\dot{H}^{1.3}}^2 dt \right)^{1/2} \left(\int_0^T \|\phi\|_{\dot{H}^{1.3}}^2 dt \right)^{1/2} \\ &\leq \varepsilon_k^{1/2} \cdot \frac{M}{\sqrt{2}} \cdot \|\phi\|_{L^2_t \dot{H}^{1.3}} \rightarrow 0. \end{aligned}$$

Step 4: Limit of the linear terms. The pressure and viscous terms pass to the limit by weak convergence.

Step 5: Energy inequality. The Leray–Hopf energy inequality for u follows from lower semicontinuity of the L^2 and H^1 norms:

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u\|_{L^2}^2 ds \leq \|u_0\|_{L^2}^2.$$

Step 6: Axisymmetric structure. The strong L^2_{loc} limit of axisymmetric-with-swirl fields is axisymmetric-with-swirl. $\square$

Remark 9.3. Theorem 9.2 produces a Leray–Hopf weak solution for axisymmetric-with-swirl data. In the without-swirl case ($u^\theta \equiv 0$), the regularity of such solutions is known [6, 13]. In the with-swirl case, regularity depends on whether $C(\varepsilon)$ in (16) is ε -independent.

10 Open Problems

OP1. (Surjectivity of Q_1 -approximation.) Does every Leray–Hopf weak solution with axisymmetric-with-swirl data in H^1 arise as the strong L^2 -limit of a sequence of NS_ε smooth solutions? This would confirm that the class studied in Theorem 9.2 is exhaustive.

OP2. (Uniform Gronwall bound — the precise gap.) The constant $C(\varepsilon)$ in (16) equals $2 \sup_t \|u_\varepsilon^r(t)/r\|_{L^\infty}$. The open condition is:

$$\sup_{\varepsilon \in (0,1]} \int_0^T \left\| \frac{u_\varepsilon^r(t)}{r} \right\|_{L^\infty} dt \leq C_0(T, M, G_0) < \infty.$$

This is a Prodi–Serrin type criterion [11] for the family $\{u_\varepsilon\}$: it asserts that the radial component is controlled near the axis uniformly in the augmentation parameter. For smooth axisymmetric fields, $u^r/r = -\partial_z u^z/2 + O(r^2)$ at the axis (by incompressibility and Taylor expansion), so the condition is equivalent to a uniform bound on $\partial_z u_\varepsilon^z$ at $r = 0$.

11 Conclusion

We have presented a complete treatment of the Q_1 -augmented axisymmetric-with-swirl Navier–Stokes system, with the following rigorously established results:

1. The substitution $\Phi = u^\theta/r$ with $\Gamma = r^2\Phi$ yields the identity $r^{-4}\partial_z(\Gamma^2) = \partial_z(\Phi^2)$ (Lemma 6.1), removing the explicit $1/r^4$ prefactor from the swirl-vorticity source and relocating the axis difficulty into the advection coefficient u^r/r .
2. The Phi-energy estimate (Theorem 6.3) closes for each fixed $\varepsilon > 0$ with constant $C(\varepsilon) = 2 \sup_t \|u_\varepsilon^r/r\|_{L^\infty}$.
3. The augmented family is uniformly bounded in $L_t^\infty L^2$ and $L_t^2 H^1$, and the Γ -bound is uniform in ε (Lemma 5.2).
4. Aubin–Lions compactness (Lemma 9.1) yields a subsequential limit that is a Leray–Hopf weak solution of the classical system (Theorem 9.2).
5. The Q_1 augmentation term vanishes in the limit via the energy bound $\varepsilon_k \int \|u_{\varepsilon_k}\|_{\dot{H}^{1.3}}^2 \leq M^2/2$ (proved in Theorem 9.2, Step 3).

The axisymmetric-with-swirl regularity problem has been reduced to a single open condition: ε -independence of the Gronwall constant $C(\varepsilon) = 2 \sup_t \|u_\varepsilon^r(t)/r\|_{L^\infty}$ (Open Problem 10).

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